

AN UNSTOPPABLE PROCESS

Alfani-Colli, chapters 6.4 and 8.2+8.3

Economic history

Ay 2025/2026

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INDUSTRIALIZATION BEYOND EUROPE: USA

- At the time of independence (1783) – The Constitution took effect in 1789
 - Almost 90% of the population was working in agriculture
 - In the following years the cotton industry in the USA became the largest in supplying raw material worldwide (1820-1860)
- The large domestic market played an important role to boost the mass production industries
- During the first half of the 19th century:
 - states, municipalities and privates invested in turnpikes and canals to integrate the huge internal market
 - Railway network in 1840 exceeded the one in all of Europe boosting transportation services and industrial sectors (iron and steel)

INDUSTRIALIZATION BEYOND EUROPE: USA

- Increasing population:
 - From 4 million in 1790 to 40 million to 1870 → **population growth**
 - **Increase in income and wealth**
 - Scarcity of labour with respect to land and other resources
→ higher wages and standard of living
 - Political and religious liberties
 - Abundance of resources explains higher per capita income but not higher rate of **economic growth** (wrt Western Europe)
 - Rapid progress technology
 - Regional specialization → favoured by high physical dimension and varied climates



THE US ECONOMY

- At the onset of the Civil War (1861-1865)
- North East specialized in manufacturing, large-scale commerce and finance;
- The West, a vast and continually expanding territory, specialised in livestock farming and the cultivation of wheat, maize and other cereals;
- The South, concentrated on colonial products (cotton, sugar cane, tobacco etc.) primarily destined for export.

INDUSTRIALIZATION BEYOND EUROPE: USA

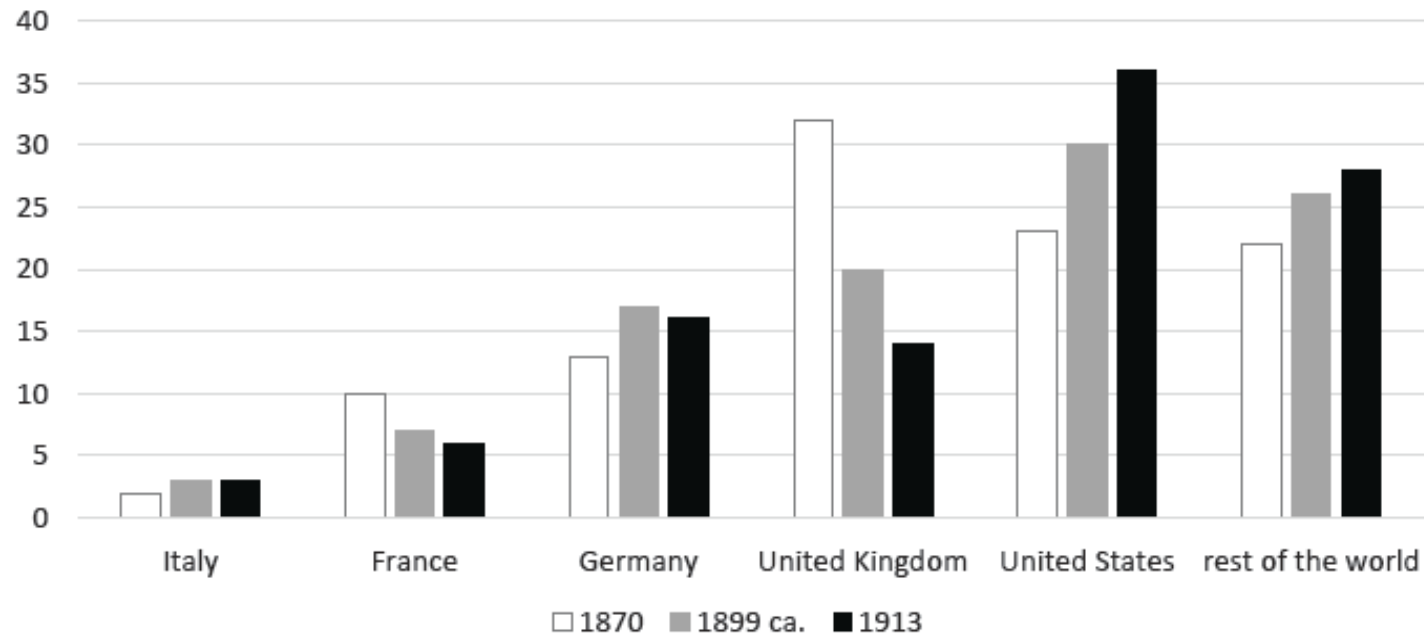
- Non agricultural labour force >workers in agriculture in 1880 and income from manufacturing began to exceed that from agriculture

→ By 1890 the USA was the most industrial nation

- Big corporations
 - Central office
 - Integration of production and distribution
 - Multinational expansion
- University education started to provide courses for mechanical, electrical and chemical engineers
- Business schools were introduced to study the strategy and management practices of the big industrial corporations

INDUSTRIALIZATION BEYOND EUROPE: USA

Figure 6.5. – *Shares of world industrial output (%)*



Source: Based on W.W. Rostow, *The World Economy: History and Prospect*, Austin 1978, p. 52.

TAYLORISM OR *SCIENTIFIC MANAGEMENT*

Mechanical engineer and business consultant, Frederick Taylor is the founding father of the scientific organisation of work (*Principles of scientific management*, 1911)

The elements that characterise it are:

- identification of the best way of doing things
- personnel selection and training
- division of labour and management tasks

Taylor's goal is to increase the **efficiency** of the production process in order to **increase profit**.

FORDISM

The Fordist method overcomes the selection and training problems of scientific management: it does not *“teach the worker the only best way to do his job”*, but *“arrange things in such a way that he can only do his best work”*.

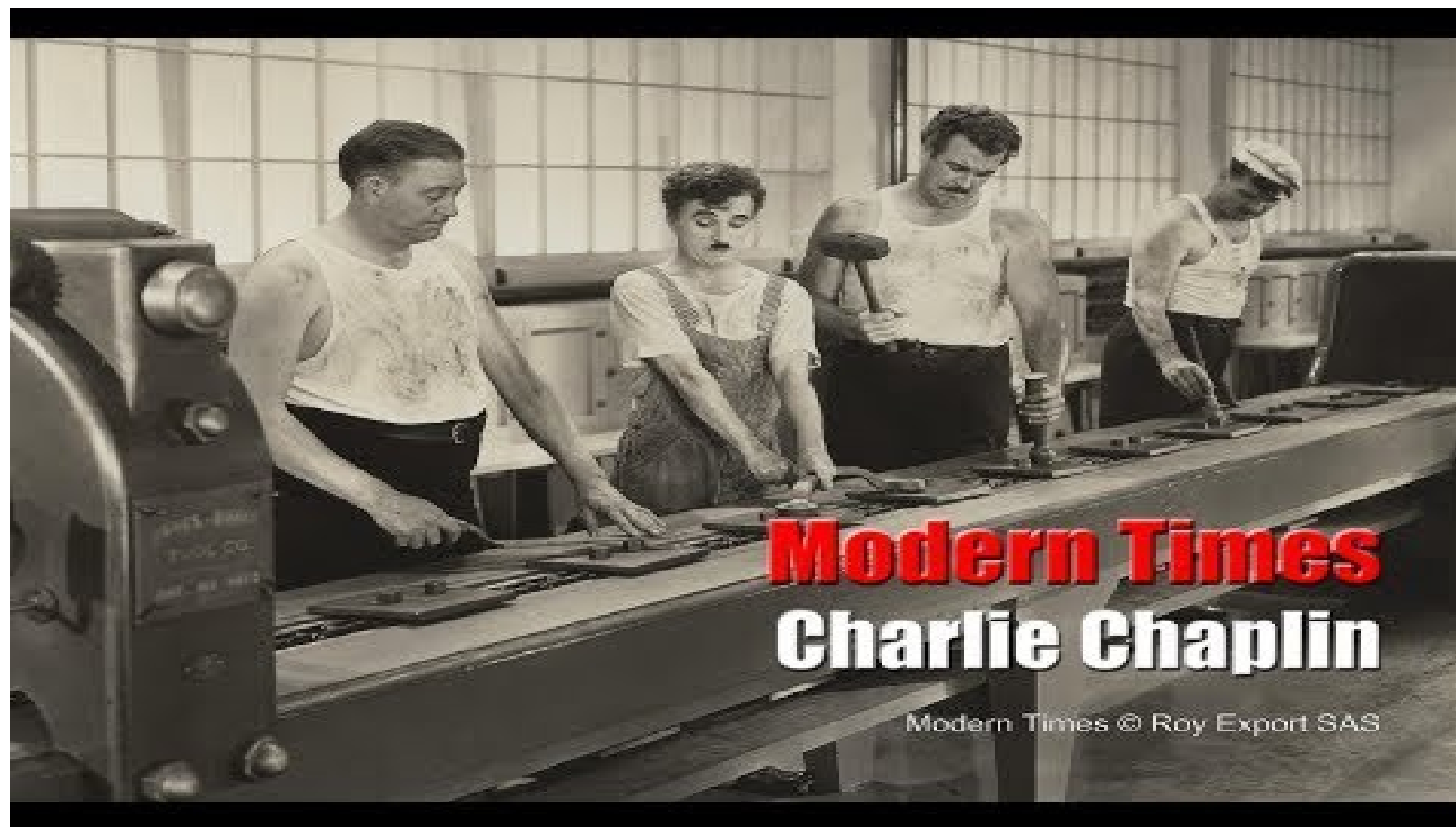
It maximises the principle of **standardisation and economies of scale**:

- The tasks of individual workers are parcelled out
- movements reduced to a minimum
- the required skills are practically non-existent

The (engineering) design of the production layout and the assembly line solve the problem of work organisation.

To this Ford combined

- high wages,
- corporate welfare, and
- total intolerance towards any form of trade union.



What about the East the relationship with the
West of the World?

INDUSTRIALIZATION BEYOND EUROPE: CHINA

- Qing Dynasty - The Manchu conquered Beijing in 1645
- Population grew from approximately circa 275 million in 1700 to approximately 350 million in 1800
- It was an agriculture economy
- Exporting mainly to Britain tea and silk
 - British products were not requested by the Chinese market

→ Silver Mexican dollars were shipped to China

- This was costly for Britain and when they discovered that Chinese had a taste for opium
 - Mexican silver substituted with Opium from Bengal
- Effects on health of China's population
- China's balance of trade became negative
- Increasing corruption and administrative disorders

INDUSTRIALIZATION BEYOND EUROPE: CHINA

- **Opium Wars** of 1840 and 1860 and the **Unequal Treaties**
 - China gave Hong Kong to Britain
 - 5 new ports were opened
 - Indemnity paid
 - Other Western Nations obtained favourable treatments
- Showing the weakness of the government
- Provoking antigovernment and antiforeign demonstrations culminating in 1900-1901 with the Boxer Rebellion
- Chinese students sent abroad for further studies, especially to Japan, became critical towards traditional society
 - Sun Yat-Sen's republican revolution of 1912 – The new Republic of China
- The political weakness prevented China to experience an early industrialisation phase

INDUSTRIALIZATION BEYOND EUROPE

- Public intervention in the process of national industrialisation is also decisive in the first examples of transformation outside (or at the margins of) Europe.
- It started at the turn of the 19th and 20th centuries, as a result of exogenous shocks and the spread of the **globalisation**.



INDUSTRIALIZATION BEYOND EUROPE: JAPAN

- Japan of the Tokugawa period (1603-1868) was a closed world (*sakoku* = chained country). Europeans arrived in Japan in 1543, but after a century no foreigners could trade, except for the Koreans, Dutch and Chinese (via Nagasaki).
- During this period, however, there were some modernising thrusts:
 - development of urban centres
 - blossoming of luxury handicrafts
 - spread of production models similar to the European putting-out system
 - formation of mercantile (rice trade) and credit circuits
 - widespread primary education
 - development of good road network: favoured rudimentary integration of the national market and the formation of regional markets





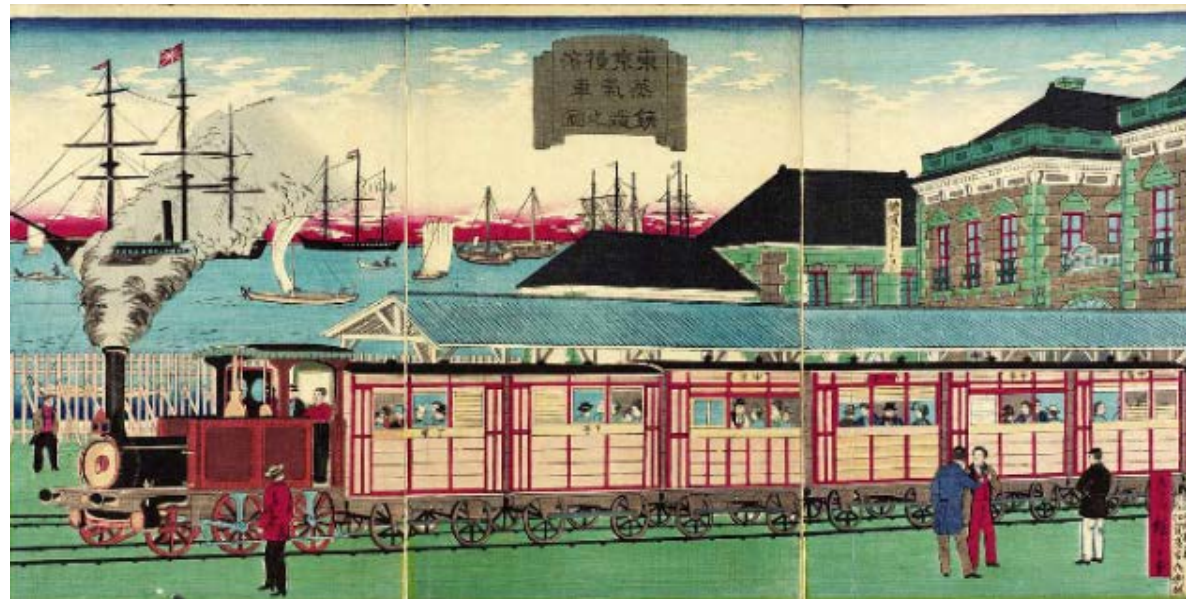


INDUSTRIALIZATION BEYOND EUROPE: JAPAN

- Japanese isolation was broken in the years 1853-1854: a US military ship entered the port of Edo (Tokyo) and forced the opening of trade relations between the two countries under threat of bombing the city.
- In short time, other countries forced Japan to enter into (unequal) commercial agreements that were a clear sign of the weakness of the shogunate:
 - the agreements granted extraterritoriality to foreign merchants
 - prevented the Japanese authorities from raising any tariff above 5%
- The opening to international trade was not beneficial to Japan: the local industries, even the more advanced (like those of cotton) were damaged by the Western competition.

INDUSTRIALIZATION BEYOND EUROPE: JAPAN

- In 1868 an alliance of aristocrats defeated the shogunate. Emperor Mutsuhito ascended the throne, to reign over Japan until 1912. 1868 is the start of the **Meiji restoration**.
- The new imperial policy was based on the goal of becoming a 'first class nation'. The slogan used was *fokoku kyohei* or 'rich nation, strong army'.



INDUSTRIALIZATION BEYOND EUROPE: JAPAN

The emperor initiated a vast modernisation programme for the country inspired by the best Western models:



- Reform of the administration
- Reform of the armed forces
- Tax reform
- Reform of the educational system
- Company law reform
- Infrastructure construction
- Modernisation of credit and banking system
- Accession to the international monetary system (gold standard)

COPYING THE WEST

- ▶ Japan imported technological knowledge from abroad.
- ▶ Employed foreign advisors and experts who worked both with the state and as employees in the private sector.
- ▶ Sent missions or individual Japanese abroad.



<i>Volume</i>	<i>Country</i>	<i>Year</i>	<i>City</i>
Volume I (Ch.1-20)	The United States of America	1872.1-8	San Francisco, Chicago, Washington D.C., Philadelphia, New York, Boston
Volume II (Ch.21-40)	Britain	1872.8-12	London, Liverpool, Manchester, Glasgow, Edinburgh, Newcastle, Bradford, Sheffield, Birmingham
Volume III (ch.41-60)	Continental Europe 1	1872.12-1873.3	Paris, The Hague, Rotterdam, Leiden, Amsterdam, Berlin
Volume IV (ch.61-81)	Continental Europe 2	1873.4-1873.6	St. Petersburg, Florence, Rome, Naples, Venice, Vienna
Volume V (ch.82-100)	Continental Europe 3, Voyage Home	1873.6-1873.9	Vienna, Berne, Geneva, Lyon, Marseilles, Hong Kong, Shanghai

List of cities visited and recorded by Iwakura Mission





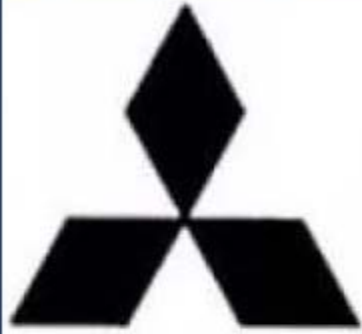
INDUSTRIALIZATION BEYOND EUROPE: JAPAN

The Meiji government tackled the problem of industrialisation in two ways:

- through the creation of infrastructures, among which the railway network had a prominent place
- through the direct establishment of production plants in the shipbuilding sector, the iron and steel sector, the production of weapons, tools and machinery, the chemical sector, sugar refining, glass, textiles

These industrial activities were then sold to private individuals, becoming the first nucleus of the large private industrial groups in formation: the *zaibatsu*.

To stimulate industrialisation the government also used the **import substitution policy**: tariff barriers on imported products to stimulate the development of domestic industry.



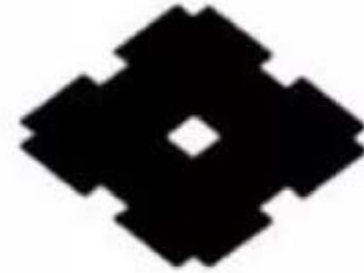
三菱

Mitsubishi



三井

Mitsui



住友

Sumitomo

Table 3–1a. Growth of Per Capita GDP, Population and GDP: World and Major Regions, 1000–1998
(annual average compound growth rates)

	<i>1000–1500</i>	<i>1500–1820</i>	<i>1820–70</i>	<i>1870–1913</i>	<i>1913–50</i>	<i>1950–73</i>	<i>1973–98</i>
Per capita GDP							
Western Europe	0.13	0.15	0.95	1.32	0.76	4.08	1.78
Western Offshoots	0.00	0.34	1.42	1.81	1.55	2.44	1.94
Japan	0.03	0.09	0.19	1.48	0.89	8.05	2.34
Asia (excluding Japan)	0.05	0.00	–0.11	0.38	–0.02	2.92	3.54
Latin America	0.01	0.15	0.10	1.81	1.42	2.52	0.99
Eastern Europe & former USSR	0.04	0.10	0.64	1.15	1.50	3.49	–1.10
Africa	–0.01	0.01	0.12	0.64	1.02	2.07	0.01
World	0.05	0.05	0.53	1.30	0.91	2.93	1.33

Source: A. Maddison, The world economy: A millennial perspective, Paris, 2007, 126